METRIC
DOD-L-24651
18 January 1985

MILITARY SPECIFICATION

LUBRICATING OIL, FOOD GRADE, FOOD PROCESSING EQUIPMENT (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of the Defense.

1. SCOPE

1.1 Scope. This specification covers lubricating oils for use in food processing equipment in applications where there may be incidental contact with food.

1.2 Classification. Lubricating oils for use in food processing equipment shall be of the following types, as specified (see 6.2):

Type I - Lubricating oil, general purpose
Type II - Lubricating oil, gears

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation and Submission of.
FED-STD-791 - Lubricants, Liquid Fuels and Related Products; Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290 - Packaging of Petroleum and Related Products.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9150

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2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein.

UNITED STATES DEPARTMENT OF AGRICULTURE (USDA)
Miscellaneous Publication No. 1419 - List of Proprietary Substances and Nonfood Compounds Authorized for Use Under USDA Inspection and Grading Programs.

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, ATTN: Mail Stop SSOM, Washington, DC 20402.)

(Copies of standards and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

D 92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup. (DoD adopted)
D 97 - Standard Test Method for Pour Point of Petroleum Oils. (DoD adopted)
D 270 - Standard Method of Sampling Petroleum and Petroleum Products. (DoD adopted)
D 445 - Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity). (DoD adopted)
D 665 - Standard Test Method for Rust Preventing Characteristics of Inhibited Mineral Oil in the Presence of Water. (DoD adopted)
D 2266 - Standard Test Method for Wear Preventive Characteristics of Lubricating Grease (Four-Ball Method). (DoD adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Material. The lubricating oil shall be a stable homogeneous blend of virgin petroleum lubricating oil stocks plus additive agents, if necessary, to meet the requirements of this specification. The finished oil shall be listed in USDA Miscellaneous Publication No. 1419, with a rating of USDA H1, lubricants approved for incidental contact with food.

3.2 Chemical and physical requirements. The lubricating oil shall conform to the chemical and physical requirements specified in table I.

TABLE I. Chemical and physical requirements.

Characteristics	Type I limits	Type II limits
Viscosity, cSt, at 40°C	28-76	195-260
Viscosity, cSt, at 100°C	5.7-8.8	15.5-22.7
Pour point (maximum)	-18°C	-12°C
Flash point (minimum)	204°C	204°C
Corrosion protection	Pass	Pass
Wear characteristics (mm maximum diameter)	0.55	0.55
USDA rating	H-1	H-1

3.3 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is form OSHA-20 and found as part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. The lubricating oil shall be inspected and tested to verify compliance with the requirements of 4.5.1.

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4.2.1 Lot.

4.2.2 Bulk lot. A bulk lot is a homogeneous mixture of one type of oil offered for acceptance in a single, isolated container (see 5.1); or manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.2.3 Packaged lot. A packaged lot shall consist of unit containers of identical size and type, offered for acceptance, and filled with a homogeneous mixture of one grade of oil from a single, isolated container (see 5.1); or filled with a homogenous mixture of one grade of oil manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.3 Sampling.

4.3.1 Sampling of filled containers. A random sample of filled containers shall be taken from each lot in accordance with MIL-STD-105, inspection level II and acceptable quality level (AQL) = 2.5 percent defective.

4.3.2 Sampling for tests. Samples shall be taken from bulk or packaged lots for tests in accordance with ASTM D 270.

4.4 Inspection. Inspection shall be in accordance with method 9601 of FED-STD-791.

4.4.1 Examination of filled containers. Samples taken in accordance with 4.3.1 shall be examined for compliance with MIL-STD-290 for fill, closure, sealing, leakage, packing, packaging and marking requirements. Containers shall also be weighed to determine the amount of the contents. Any container having one or more defects or under required fill, shall be rejected. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 Test methods. Tests for the requirements specified in 3.2 shall be in accordance with table II.

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TABLE II. Chemical and physical test requirements.

Characteristics	Test reference
Viscosity, cSt, at 40°C	ASTM D 445
Viscosity, cSt, at 100°C	ASTM D 445
Pour point (maximum)	ASTM D 97
Flash point (minimum)	ASTM D 92
Corrosion protection	ASTM D 665
Wear characteristics (mm maximum diameter)	ASTM D 2266
USDA rating	See 4.5.2

4.5.2 USDA rating. The lubricating oil shall be listed in USDA Miscellaneous Publication No. 1419, and rated USDA H1, lubricants approved for incidental contact with food.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging, packing and marking. Unless otherwise specified (see 6.2), packaging, packing and marking shall be in accordance with MIL-STD-290. The size of the containers shall be 18.9 liter (5 gallon) cans unless otherwise specified (see 6.2).

6. NOTES

6.1 Intended use. The oils covered by this specification are intended for use as a lubricant in food processing equipment.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of oil (see 1.2).
(c) Packaging, packing and marking, if other than specified (see 5.1).
(d) If other than 18.9 liter (5 gallon) cans are required (see 5.1).

Custodians:
Navy - SH
Army - ME
Air Force - 68

Review activity:
DLA - GS

Preparing activity:
Navy - SH
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1. DOCUMENT NUMBER	2 DOCUMENT TITLE
DOD-L-24651	

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
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5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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